

# Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 1: Meeting the Phenomenon: Mr. Wafula's Maize and Beans

Teacher reference only. Do not project or hand out.

## Quick marking grid

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 |
|----|----|----|----|----|----|----|
| C  | C  | D  | B  | C  | D  | B  |

USE AFTER: OBSERVE PHASE — Farm data: Day 14 germination counts (maize 88/100, beans 80/100)

**Q1. On Mr. Wafula's farm near Kitale, 100 bean seeds were planted. By Day 14, 80 of them had germinated. How many bean seeds had NOT germinated by Day 14?**

- A. 8
- B. 80
- C. 20**
- D. 12

**Correct answer: C.** The seeds that did not germinate are the total minus those that germinated:  $100 - 80 = 20$ . The figure 12 is the count for maize (88 germinated), not beans.

USE AFTER: OBSERVE PHASE — Watching the two time-lapse videos and completing the events table (radicle, emergence, first leaf, branching)

**Q2. In the time-lapse records, the bean radicle appeared on Day 3 and the bean shoot came above the soil on Day 8. The maize radicle appeared on Day 4 and the maize shoot came above the soil on Day 7. Which statement matches these records?**

- A. Maize was first for both the radicle and the shoot above soil
- B. Both crops had their radicle and shoot above soil on the same days
- C. The bean radicle appeared first, but the maize shoot came above the soil first**
- D. Bean was first for both the radicle and the shoot above soil

**Correct answer: C.** Bean radicle Day 3 comes before maize radicle Day 4, but maize emergence Day 7 comes before bean emergence Day 8. So the order is different for the two events.

USE AFTER: OBSERVE PHASE — Oral probing: 'How do you know the bean germinated before it emerged?'

**Q3. A group is studying the bean time-lapse. Their table shows the radicle appeared on Day 3 and the shoot came above the soil on Day 8. What can they say about the bean seed on Day 5?**

- A. It had not germinated yet because no shoot was above the soil
- B. It had failed to germinate because nothing was visible above the soil
- C. It had both germinated and emerged
- D. It had germinated because the radicle had appeared, but it had not yet emerged above the soil**

**Correct answer: D.** Germination was recorded when the radicle appeared (Day 3). Emergence was recorded when the shoot came above the soil (Day 8). On Day 5 the first had happened but not the second.

USE AFTER: OBSERVE PHASE — Studying the farm data and completing 'I notice that...' statements (stems also thicken and branch)

**Q4. Learners watched a bean plant for 25 days. Which observation shows that plant growth is more than just getting taller?**

- A. Maize was taller than beans from Day 14 onward
- B. The stem became thicker and the bean plant developed branches by Day 25**
- C. More maize seeds than bean seeds germinated by Day 14
- D. The bean radicle appeared on Day 3

**Correct answer: B.** Thickening of the stem and branching are changes other than an increase in height. The other options describe height, germination counts or timing of an event.

USE AFTER: EXPLAIN PHASE — Initial model and sentence starter: 'I think some healthy-looking seeds do not germinate because...'

**Q5. Mr. Wafula planted maize and beans on the same day, 3 cm deep, after 25 mm of rain, in the same field. Some healthy-looking seeds did not germinate. At the end of this first lesson, which statement is the most honest one?**

- A. Healthy-looking seeds that fail must have been damaged by pests
- B. The seeds failed because they were planted too deep
- C. We do not yet know the reason; it is a question we need to investigate**
- D. The seeds failed because beans need more water than maize

**Correct answer: C.** The lesson only records observations and possible factors to test, and the cause has not been investigated yet. The other options state causes that the evidence collected so far does not support.

USE AFTER: DQB PHASE — Driving Question Board: groups write questions on sticky notes or cards

**Q6. Groups are writing questions for the Driving Question Board about Mr. Wafula's maize and beans. Which question is written in a way that could be investigated?**

- A. Is farming a hard job?
- B. Why is maize better than beans?
- C. Who planted the seeds first?
- D. Does the amount of water given to bean seeds change how many of them germinate by Day 14?**

**Correct answer: D.** This question names something that can be changed (the amount of water) and something that can be measured (the number of seeds germinating by Day 14). The other questions are vague, opinion-based or not about growth.

USE AFTER: MODEL BUILDING PHASE — Exit ticket: 'What did we observe today?'

**Q7. For an exit ticket, a learner must write an observation, not an explanation, about Mr. Wafula's crops. Which statement is an observation?**

- A. Some seeds failed because the bean seed has less food inside
- B. Maize emerged above the soil on Day 7 and beans on Day 8**
- C. Maize emerged first because it is stronger than beans
- D. Beans need more water to germinate

**Correct answer: B.** An observation records what was seen or measured, such as the day of emergence. The other options give reasons that have not been tested.